IMEN574 / IDSC574 · PROGRAMMING FOR AX

# 13

# Interpretable ML

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# Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

# Outline: Interpretable ML

01 **Interpretation Targets**

Define the question

02 **Feature Effects**

Inspect predictive reliance

03 **Local and Example Evidence**

Validate the explanation

# 01 · Interpretation Targets

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Define the question

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# Interpretation Questions

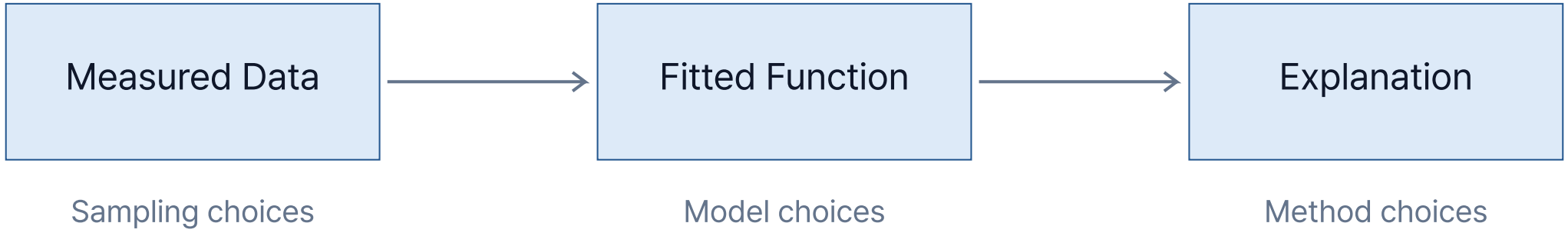
Choose whether to explain a model or one prediction

| Scope | Question | Example |
| --- | --- | --- |
| Global | What does the model use? | Permutation importance |
| Local | Why this output? | SHAP, LIME |
| Action | What could change? | Counterfactual |

The question determines which evidence is relevant

# Model and World

An explanation describes a fitted function under assumptions




An explanation inherits the model and data limitations

# Transparent Model Structure

A simple model exposes its prediction calculation

| | Value | Coefficient | Contribution |
|---|---|---|---|
| Feature A | 2 | 2 | 4 |
| Feature B | 4 | 1 | 4 |
| Intercept | 0 | 1 | 0 |

$$\hat{y} = 2a + b$$

The weighted terms sum to the prediction

# Coefficient Scale

Feature units change the numerical coefficient

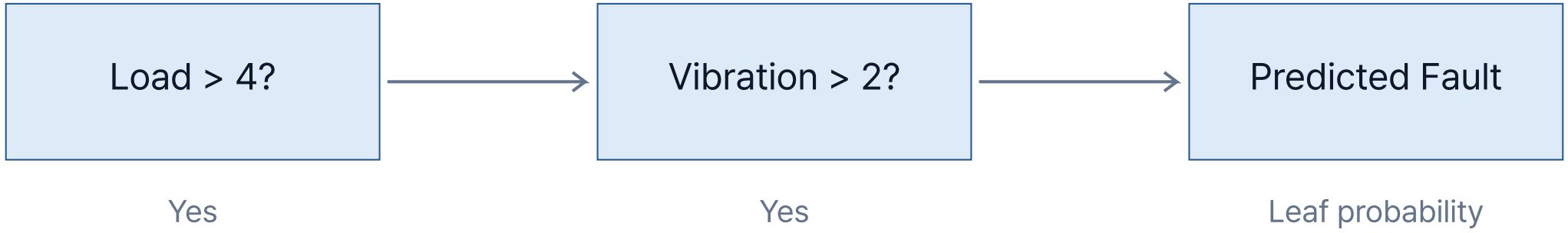
| Unit | Coefficient |
| --- | --- |
| Meters | 2 |
| Centimeters | 0.02 |

$$2 \times 1\ \mathrm{m} = 0.02 \times 100\ \mathrm{cm}$$

The prediction is unchanged although the coefficient differs

# Interpretable Trees

A short decision path exposes the applied conditions




A path names the conditions used for this input

# Accuracy and Interpretability

Treat the tradeoff as an empirical question

| Candidate | Evaluate | Explain |
| --- | --- | --- |
| Linear model | Same held-out split | Coefficients |
| Small tree | Same held-out split | Decision paths |
| Ensemble | Same held-out split | Post-hoc methods |

Compare actual models before assuming complexity is necessary

# Explanation Taxonomy

Methods differ by access and scope

| | Global | Local |
|---|---|---|
| Model-specific | Coefficients | Tree path |
| Model-agnostic | Permutation | SHAP or LIME |

Model access and explanation scope are separate axes

# A Known Test Function

Use a small function to verify explanation arithmetic

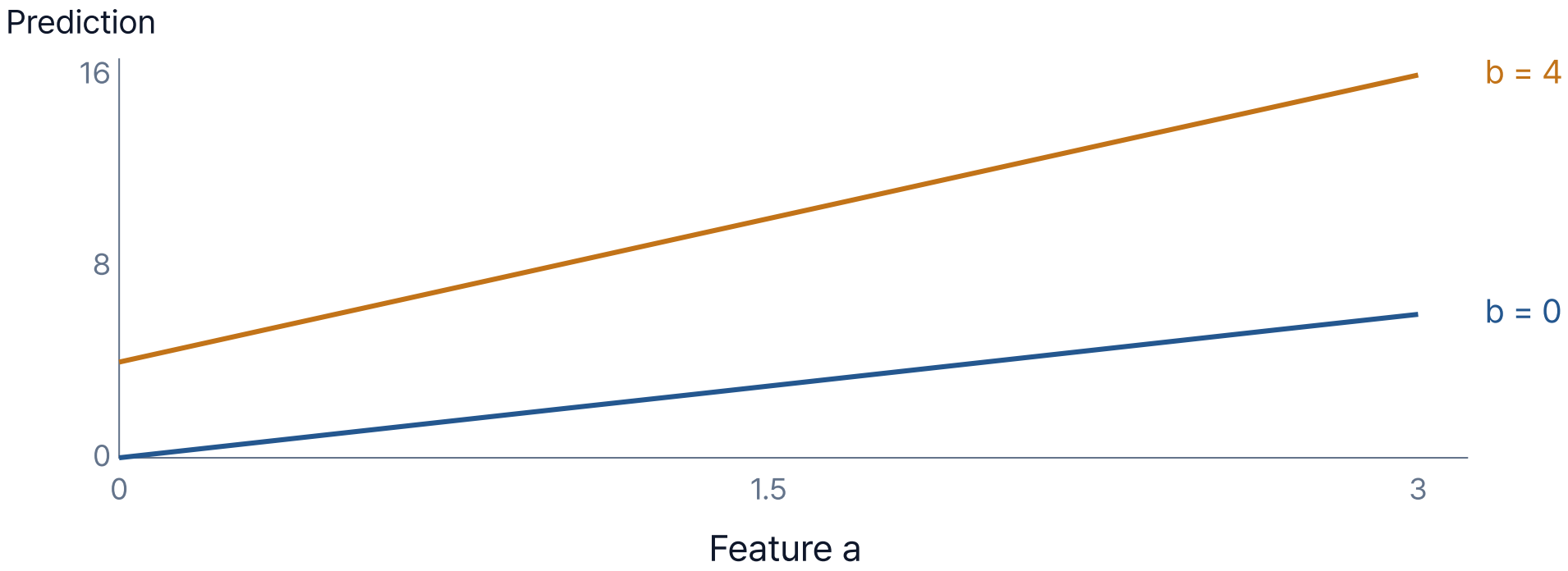
| a | b | f(a,b) |
|---|---|---|
| 2 | 4 | 12 |
| 1 | 4 | 8 |
| 2 | 0 | 4 |

$$f(a,b) = 2a + b + 0.5ab$$

The interaction makes feature effects depend on the other value

# Feature Interactions

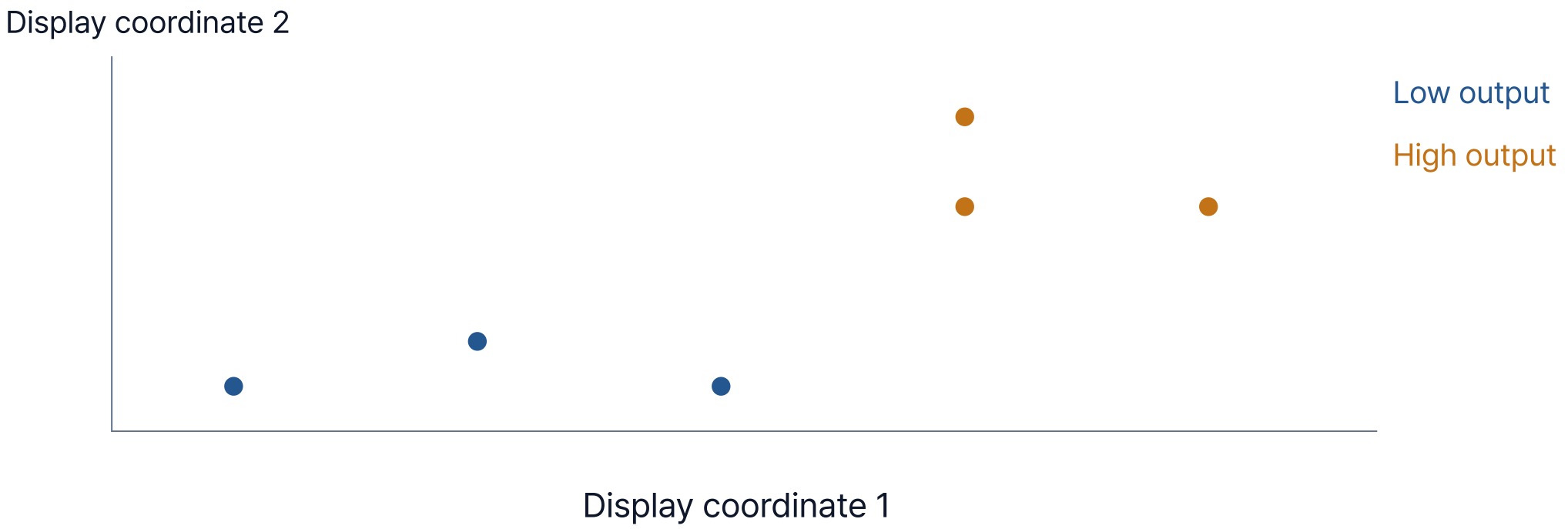
The effect of a changes with b




The slope increases from two to four as b changes

# Visualization as Evidence

A projection suggests patterns without explaining the predictor




Colored embedding groups motivate a check rather than a causal claim

# Explanation Quality Criteria

Check fidelity, stability, and relevance separately

| Criterion | Check |
| --- | --- |
| Fidelity | Does the explanation track this model? |
| Stability | Do small changes alter the result? |
| Relevance | Does it answer the user question? |
| Validity | Are perturbed inputs plausible? |

A readable explanation can still fail fidelity or validity

# 02 · Feature Effects

01 Interpretation Targets

Define the question

02 **Feature Effects**

Inspect predictive reliance

03 Local and Example Evidence

Validate the explanation

# Permutation Importance

Break one feature relation and measure the score change




Held-out data

Other columns fixed

Compare with baseline

The test measures predictive reliance under a chosen perturbation

# Permutation Arithmetic

Compute the loss increase for one exact shuffle

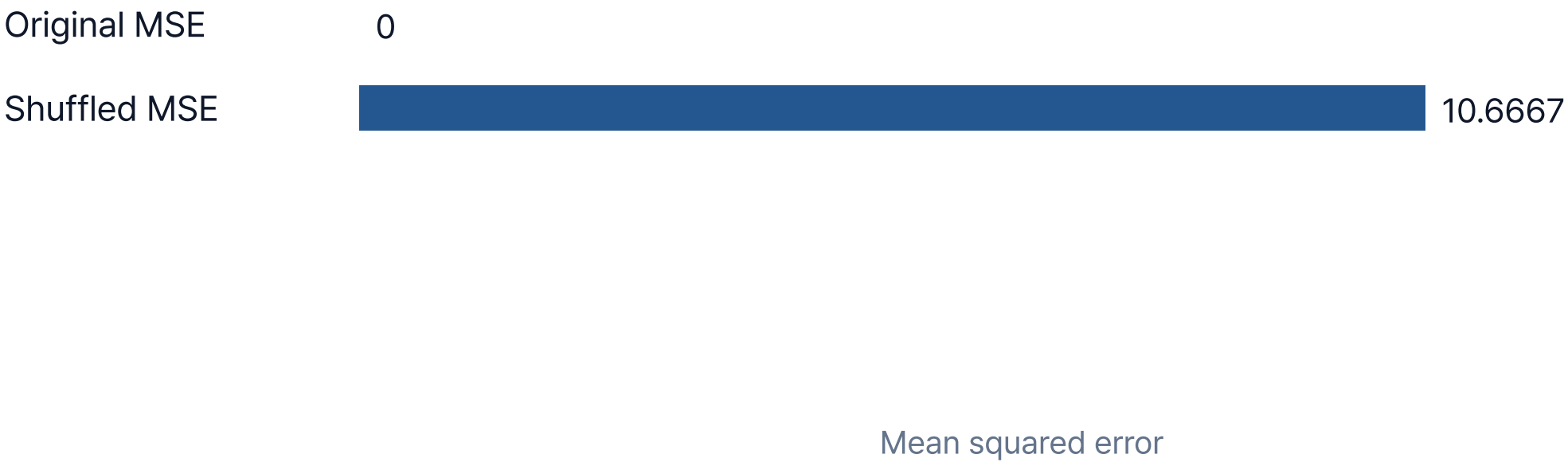
| a | True y | Model 2a |
|---|---|---|
| 0 | 0 | 0 |
| 1 | 2 | 2 |
| 2 | 4 | 4 |

```
Original MSE = 0
```

The fitted rule exactly predicts this constructed target

# A Broken Feature Relation

Reverse the feature values while keeping labels fixed




The chosen shuffle increases MSE by 10.67

# Caveat: Correlated Predictors

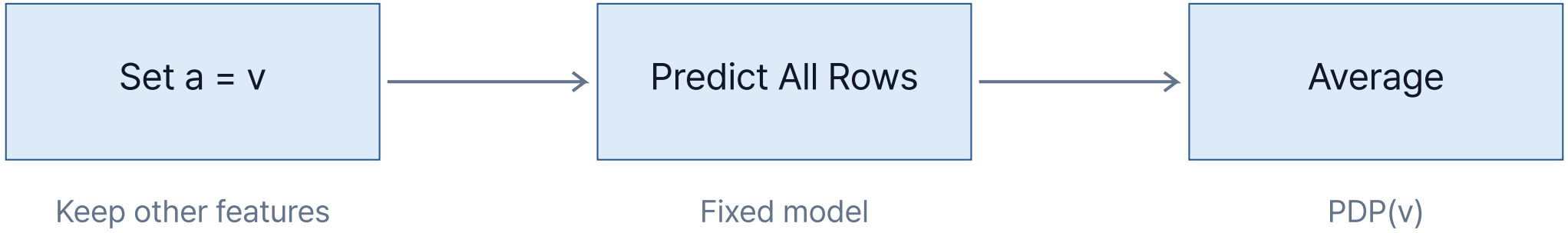
A substitute feature can hide individual reliance

| a | Copy of a |
|---|---|
| 1 | 1 |
| 2 | 2 |
| 3 | 3 |

Perfectly duplicated columns provide interchangeable evidence

# Partial Dependence

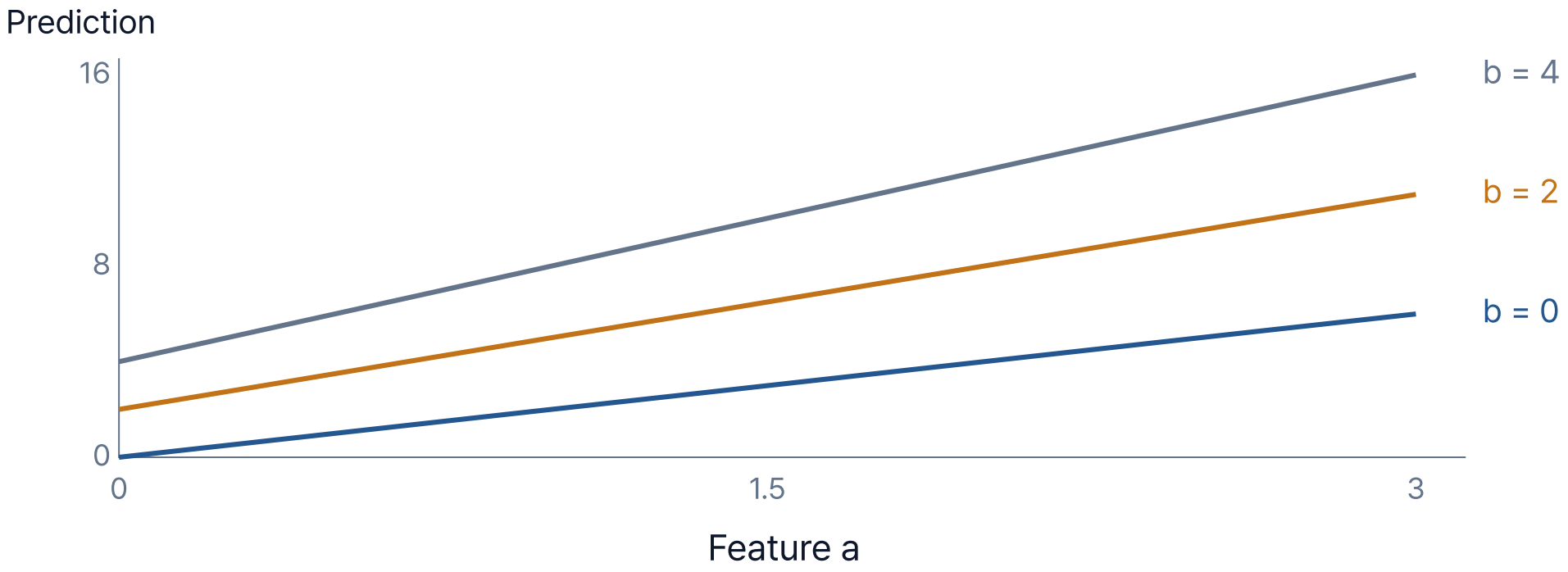
Average predictions after replacing one feature value




The average effect depends on the background rows

# Individual Conditional Expectations

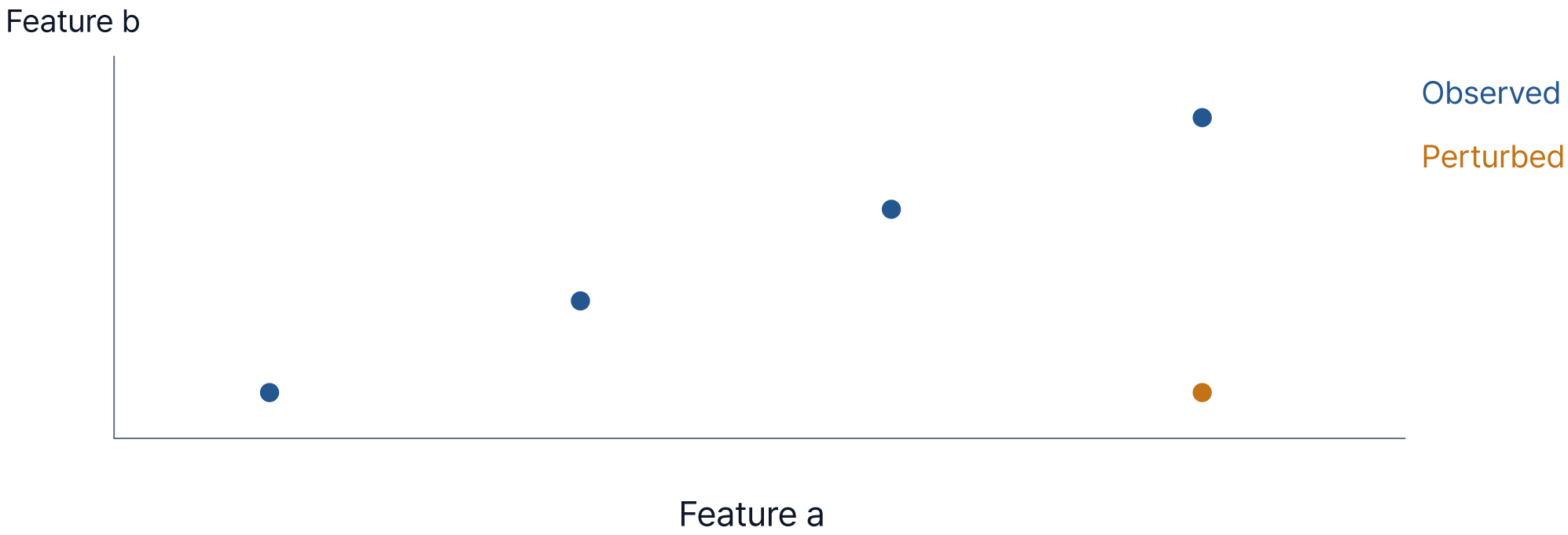
Retain separate curves before averaging




Different slopes reveal interaction with b

# Caveat: Unsupported Combinations

Independent perturbation can leave the observed data manifold




The perturbed point is far from the observed relationship

# Shapley Attribution

Average a feature contribution over feature orderings

| Ordering | a Contribution | b Contribution |
|---|---|---|
| a then b | 4 | 8 |
| b then a | 8 | 4 |
| Average | 6 | 6 |

The two-feature interaction is shared across both orderings

# Additive Explanation Check

Attributions must add to the output relative to a baseline

| | Baseline | φa | φb | Prediction |
|---|---|---|---|---|
| Known Function | 0 | 6 | 6 | 12 |

$$f(x) = \text{baseline} + \Sigma_j \, \varphi_j$$

The contributions sum exactly to twelve in this example

# Baseline Dependence

Change the reference population and change the question

| Background Point | Model Output |
| --- | --- |
| Reference (0,0) | 0 |
| Reference (1,1) | 3.5 |

The output difference depends on the reference state

# Local Attribution Magnitudes

A local magnitude describes one selected input




Attribution magnitude at the chosen input

Equal magnitude does not establish equal causal importance

# 03 · Local and Example Evidence

01 Interpretation Targets

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02 Feature Effects

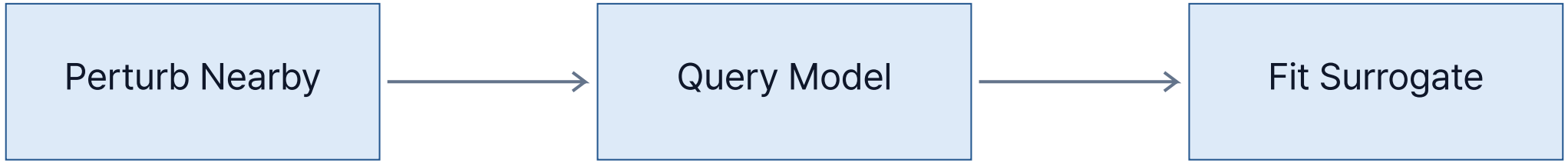
Inspect predictive reliance

03 **Local and Example Evidence**

Validate the explanation

# LIME Surrogate

Fit a simple local approximation around one input




Neighborhood choice

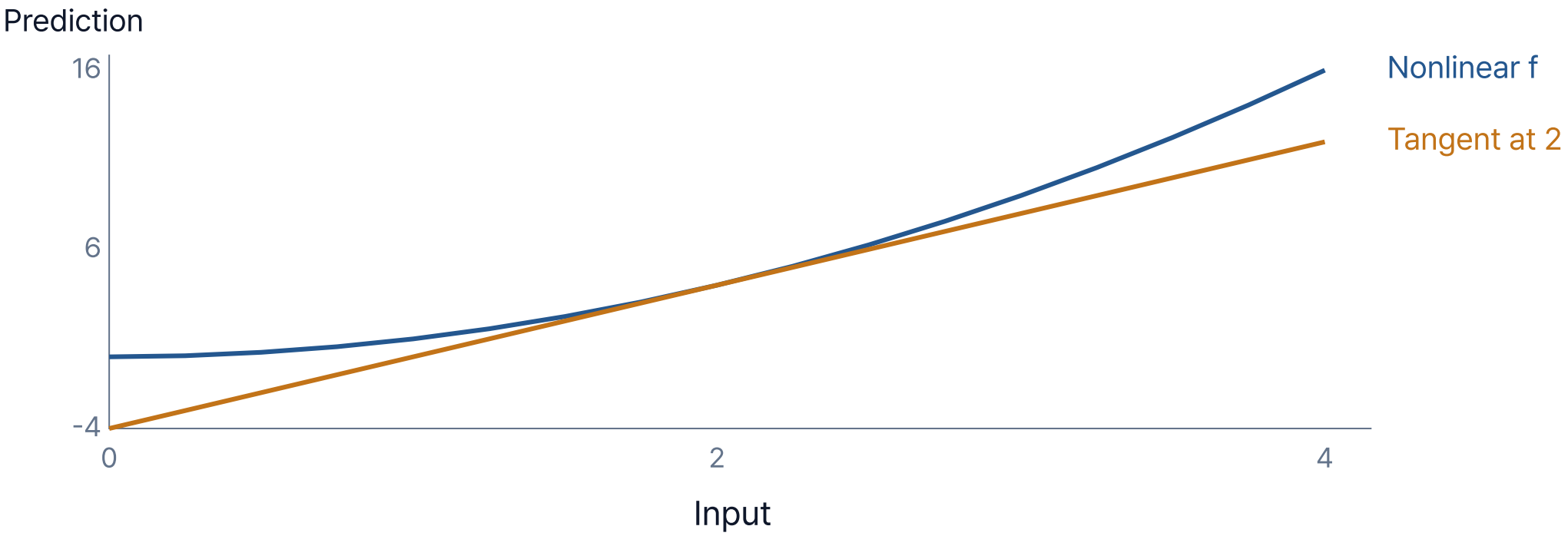
Outputs

Weighted simple model

A local model approximates the original near one selected input

# Local Fidelity

An approximation can be accurate nearby and poor far away




The tangent agrees locally but separates away from two

# SHAP and LIME Questions

Choose attribution or local approximation deliberately

| Method | Primary Construction | Check |
| --- | --- | --- |
| SHAP | Coalition contribution | Background and assumptions |
| LIME | Local surrogate | Neighborhood fidelity |

Different constructions can provide different explanations

# Counterfactual Explanations

Search for a plausible input that changes the decision

| State | a | b | Prediction |
|---|---|---|---|
| Current | 2 | 4 | 12 |
| Changed | 1 | 4 | 8 |

Reducing a changes the constructed model output from twelve to eight

# Feasible Counterfactuals

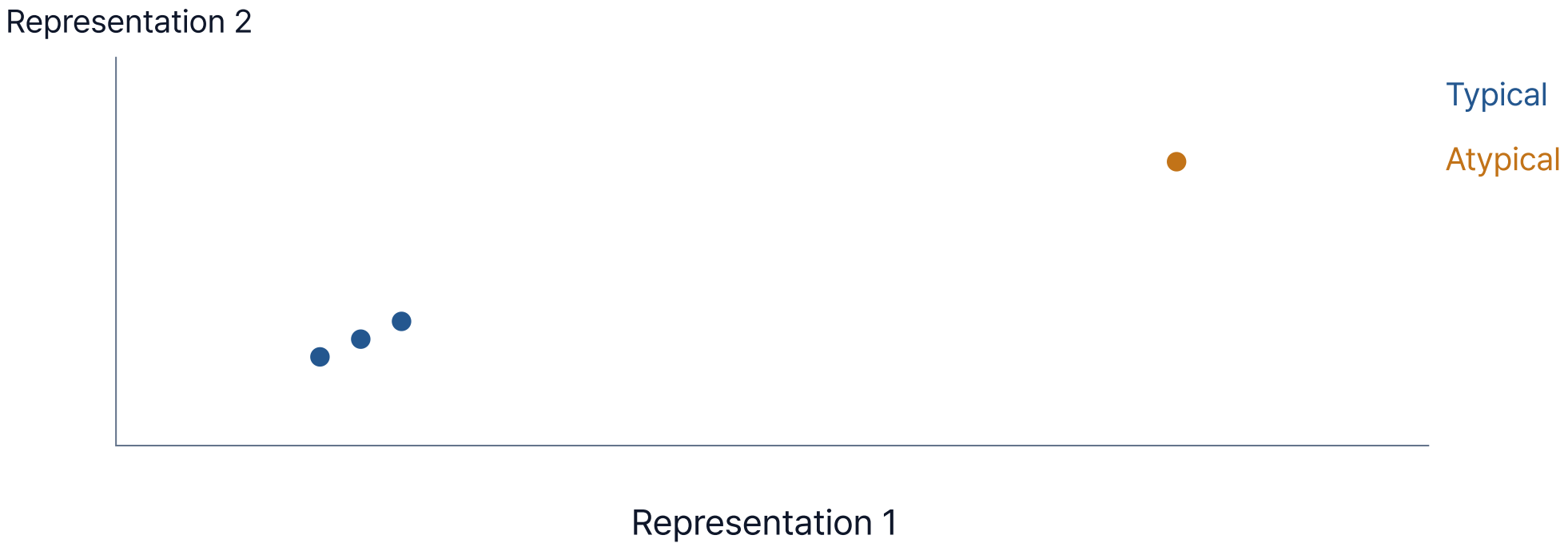
Restrict changes to actionable and valid features

| Feature | Constraint |
| --- | --- |
| Age | Fixed |
| Machine setting | Bounded |
| Unit ID | Fixed |
| Material mix | Feasible combinations |

A mathematical edit may not be a feasible action

# Prototypes and Criticisms

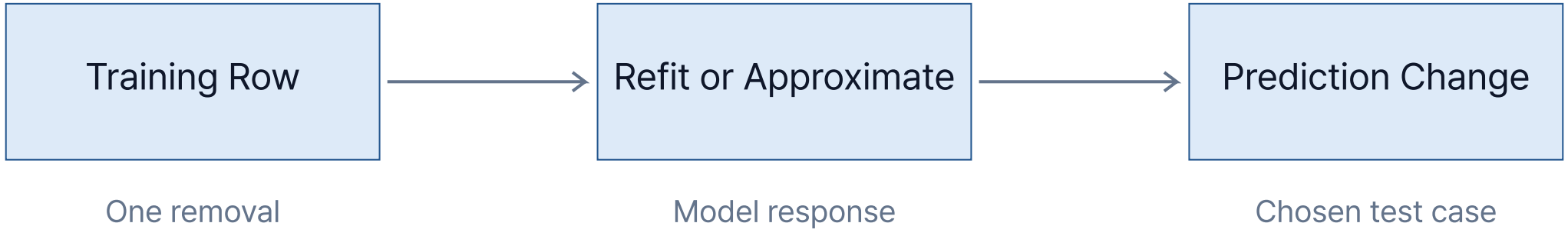
Show representative cases and poorly represented cases




An atypical case exposes what a prototype omits

# Training-Example Influence

Ask how a training observation affects the fitted result




Influence concerns fitting rather than feature perturbation

# Exact Two-Feature Attributions

Verify the coalition arithmetic without a package

```
def f(a,b): return 2*a+b+.5*a*b
a,b=2.,4.
pa=.5*((f(a,0)-f(0,0))+(f(a,b)-f(0,b)))
pb=.5*((f(0,b)-f(0,0))+(f(a,b)-f(a,0)))
print("a",pa,"b",pb,"sum",pa+pb)
assert np.isclose(pa+pb, f(a,b)-f(0,0))
```

```
a 6.0 b 6.0 sum 12.0
```

# Caveat: Explanations and Fairness

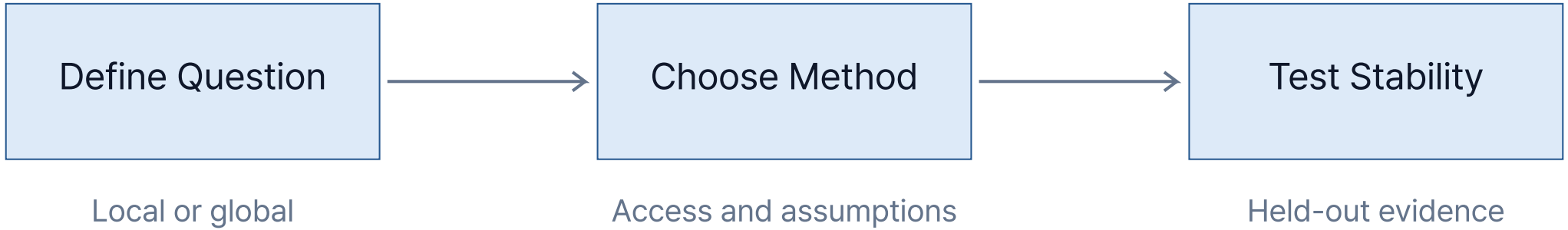
Predictive attribution does not prove fair treatment

| Claim | Additional Evidence |
| --- | --- |
| Uses a sensitive attribute | Model and input audit |
| Has disparate errors | Group-specific evaluation |
| Causes unfair outcomes | Decision process and causal analysis |

Different fairness claims require different evidence

# An Explanation Audit

Keep the question, model, and perturbation aligned




A trustworthy explanation includes its reference and validity checks

# Takeaway: Interpretable ML

01 Explain a chosen model under explicit assumptions

02 Check perturbation validity and reference data

03 Separate explanation from causation and fairness claims